

CLEVELAND HOPKINS, OH, USA

WMO: 725240

Lat: 41.406N

Lon: 81.852W

Elev: 781

StdP: 14.29

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14820

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.6	9.0	-4.5	4.4	5.5	0.3	5.7	10.5	28.6	29.4	26.4	27.7	10.6	230	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	89.7	73.6	87.2	72.4	84.6	71.1	76.2	85.5	74.7	83.3	73.2	81.4	11.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	126.9	81.4	71.8	121.1	79.9	70.4	115.2	78.5	40.0	85.9	38.6	83.2	37.2	81.8	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	21.1	19.0	-1.4	93.1	7.2	2.5	-6.6	94.9	-10.8	96.4	-14.9	97.8	-20.1	99.6
			-2.2	78.8	7.1	2.0	-7.2	80.3	-11.4	81.4	-15.3	82.6	-20.5	84.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	27.7	30.1	37.9	49.7	60.4	69.8	73.5	72.2	65.8	54.5	43.2	33.4	
	DBStd	18.50	11.83	11.31	11.05	9.67	8.81	6.95	5.47	5.21	7.14	8.50	9.29	9.88	
	HDD50	2616	697	563	406	123	12	0	0	0	1	47	246	521	
	HDD65	5737	1157	976	842	468	201	33	4	6	75	341	655	979	
	CDD50	3207	5	7	32	114	333	593	728	687	473	187	41	7	
	CDD65	853	0	0	2	9	57	176	266	228	98	17	0	0	
	CDH74	6744	0	0	10	92	499	1418	2234	1667	715	107	2	0	
	CDH80	1965	0	0	0	12	122	443	717	455	201	15	0	0	
Wind	WSAvg	9.5	11.4	10.8	10.7	10.5	9.1	8.3	7.8	7.5	8.0	9.3	10.4	10.7	
Precipitation	PrecAvg	36.9	2.2	2.2	3.0	3.2	3.6	3.5	3.5	3.4	3.5	2.6	3.2	3.0	
	PrecMax	53.8	4.4	4.7	5.2	6.6	9.1	9.1	9.1	9.0	7.3	5.6	8.8	8.6	
	PrecMin	18.8	0.4	0.5	0.9	1.3	1.0	0.6	0.7	0.6	0.7	0.7	0.8	1.1	
	PrecStd	6.3	1.0	1.0	1.0	1.2	1.7	1.8	1.5	1.8	1.5	1.1	1.7	1.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.8	66.7	74.5	82.0	87.0	91.3	93.0	91.3	90.4	81.6	71.0	63.2	
		MCWB	56.1	57.0	60.4	64.5	69.7	73.1	76.4	75.4	72.8	68.4	59.6	56.9	
	2%	DB	56.4	58.0	67.4	76.5	83.5	88.4	89.7	88.0	85.5	77.0	65.5	57.7	
		MCWB	52.4	51.3	56.4	61.2	68.5	72.4	74.3	73.2	71.2	65.2	57.4	52.5	
	5%	DB	50.5	51.5	61.9	71.5	80.1	85.4	86.9	84.9	81.4	72.7	61.7	52.4	
		MCWB	46.2	45.7	53.7	58.4	66.8	71.2	72.8	72.0	69.0	62.6	54.7	48.2	
	10%	DB	44.1	45.8	55.9	66.4	75.7	81.9	84.0	82.1	77.8	68.4	57.8	47.2	
		MCWB	40.1	41.1	48.9	55.7	64.5	69.5	71.3	70.7	67.3	60.5	51.2	43.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.9	58.6	62.0	67.6	73.2	76.4	79.0	77.8	75.4	71.0	63.1	58.4
			MCDB	60.6	66.1	72.4	76.8	83.6	86.7	89.7	87.3	85.6	79.6	68.7	62.2
2%		WB	52.9	52.3	58.1	63.5	70.6	74.4	76.7	75.5	73.1	67.6	59.0	53.6	
		MCDB	56.5	56.8	65.5	73.5	80.3	84.5	86.6	83.5	81.7	74.0	64.4	56.9	
5%		WB	46.9	46.3	54.3	60.1	68.4	72.8	74.9	74.1	71.1	64.2	55.5	48.4	
		MCDB	50.0	50.8	60.9	69.2	77.2	82.4	83.3	81.6	78.2	70.7	60.7	51.7	
10%		WB	40.4	41.1	49.5	56.9	66.0	71.2	73.2	72.5	69.1	61.3	51.8	43.6	
		MCDB	43.9	45.9	56.3	65.4	74.1	79.8	81.1	79.6	75.9	67.4	56.5	47.1	
Mean Daily Temperature Range	5% DB	MDBR	12.8	13.8	15.3	18.0	18.1	17.1	16.8	16.5	17.1	15.9	13.6	11.5	
		MCDBR	18.3	21.3	24.0	25.0	22.5	20.4	19.6	19.5	21.4	21.1	18.8	17.1	
		MCWBR	16.0	17.0	15.9	15.5	12.1	9.5	8.8	9.0	10.3	12.1	13.8	14.4	
	5% WB	MCDBR	18.0	20.1	22.0	22.7	19.8	18.3	16.9	16.4	16.9	17.9	17.4	17.0	
		MCWBR	16.6	17.5	16.0	15.8	12.0	9.8	8.5	8.6	9.2	11.6	14.4	15.8	
Clear-Sky Solar Irradiance	taub	0.305	0.335	0.357	0.395	0.431	0.449	0.447	0.433	0.385	0.367	0.326	0.309		
	taud	2.403	2.279	2.297	2.278	2.213	2.208	2.234	2.260	2.370	2.399	2.483	2.469		
	Ebn at Noon	268	274	281	278	269	264	263	263	269	260	259	258		
	Edh at Noon	25	34	37	41	45	45	43	41	34	30	24	22		
All-Sky Solar Radiation	RadAvg	437	710	1045	1424	1706	1890	1897	1680	1342	861	544	358		
	RadStd	40	92	102	122	146	139	119	76	121	74	85	36		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+266	+157
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	N/A

Nomenclature: See separate page